



Review article

Associations of environmental factors with neurodegeneration: An exposome-wide Mendelian randomization investigation

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ABSTRACT

Neurodegenerative diseases (NDDs) remain a global health challenge. Previous studies have reported potential links between environmental factors and NDDs, however, findings remain controversial across studies and elusive to be interpreted as evidence of robust causal associations. In this study, we comprehensively explored the causal associations of the common environmental factors with major NDDs including Alzheimer's disease (AD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), and multiple sclerosis (MS), based on updated large-scale genome-wide association study data through two-sample Mendelian randomization (MR) approach. Our results indicated that, overall, 28 significant sets of exposure-outcome causal association evidence were detected, 12 of which were previously underestimated and newly identified, including average weekly beer plus cider intake, strenuous sports or other exercises, diastolic blood pressure, and body fat percentage with AD, alcohol intake frequency with PD, apolipoprotein B, systolic blood pressure, and forced expiratory volume in 1 s (FEV₁) with ALS, and alcohol intake frequency, hip circumference, forced vital capacity, and FEV₁ with MS. Moreover, the causal effects of several environmental factors on NDDs were found to overlap. From a triangulation perspective, our investigation provided insights into understanding the associations of environmental factors with NDDs, providing causality-oriented evidence to establish the risk profile of NDDs.

Abbreviations: AD,, Alzheimer's disease; ADGC,, Alzheimer Disease Genetics Consortium; ALS,, Amyotrophic lateral sclerosis; BMI,, Body mass index; CHARGE,, Cohorts for Heart and Aging Research in Genomic Epidemiology Consortium; CI,, Confidence interval; CRP,, C-reactive protein; DBP,, Diastolic blood pressure; EADI,, European Alzheimer's Disease Initiative; FEV₁,, Forced expiratory volume in 1 s; FVC,, Forced vital capacity; GERAD/PERADES,, Genetic and Environmental Risk in AD/Defining Genetic, Polygenic and Environmental Risk for Alzheimer's Disease Consortium; GWASs,, Genome-wide association studies; IL,, Interleukin; IMSCG,, International Multiple Sclerosis Genetics Consortium; IPDGC,, International Parkinson's Disease Genomics Consortium; IVs,, Instrumental variables; IVW,, Inverse variance weighted; LD,, Linkage disequilibrium; LDL,, Low-density lipoprotein; MAF,, Minor allele frequency; MR,, Mendelian randomization; MR-PRESSO,, MR Pleiotropy Residual Sum and Outlier; MS,, Multiple sclerosis; MVPA,, Moderate to vigorous physical activity; NDDs,, Neurodegenerative diseases; OR,, Odds ratio; PD,, Parkinson's disease; RCTs,, Randomized controlled trials; SBP,, Systolic blood pressure; SNPs,, Single nucleotide polymorphisms; SOD1,, Superoxide dismutase 1 protein; SSOE,, Strenuous sports or other exercises; VEGFR1,, Vascular endothelial growth factor receptor 1; WM,, Weighted median; 25OHD,, 25 hydroxyvitamin D.

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1. Introduction

Achieving healthy aging remains a tremendous challenge (WHO, 2019). Although life expectancy has increased globally with the full support of medical and health services, neurodegenerative diseases (NDDs), the main neurological disorders, are expected to rise as a major burden falling on the elderly population (Hansson, 2021; Partridge et al., 2018). According to the 2017 Global Burden of Disease Study, neurological disorders account for the highest percentage change in all-age years of life lost and have presented enormous threat and are responsible for the largest percentage change of deaths in the older age among non-communicable diseases over the past decade (GBD 2017 Causes of Death Collaborators, 2017).

NDDs are characterized by the disruption in connectivity and communication between neurons and the progressive functional loss in the central or peripheral nervous system, resulting in impairment of motor and cognitive processes. The major types of NDDs include Alzheimer's disease (AD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), and multiple sclerosis (MS) (Fang et al., 2020). Although genetic susceptibilities are the major risk factors contributing to the development of these NDDs with discrete morphological and pathophysiological features, studies have indicated that several shared prodromal mechanisms including deposition of insoluble protein aggregations, neuroinflammation, metabolic stress, and neurovascular coupling potentially drive the development of NDDs, bridge the commonality of multiple NDDs, and participate in the long-term underlying impacts of environmental factors experienced throughout life on the onset, progression, and severity of NDDs (De Jager and Bennett, 2013; Fang et al., 2020; Gan et al., 2018; Wareham et al., 2022). Therefore, in the context of the current lack of effective clinical treatments for NDDs, targeting early interventions on environmental factors to improve general health is a promising strategy (Livingston et al., 2020; Wareham et al., 2022). Of note, in this investigation, the "environmental factors" is a general concept defined as non-genetic, exogenous factors that include both modifiable and non-modifiable factors in individuals (Ahn and Hayes, 2021).

Mounting evidence suggests that the environmental factors associated with the major NDDs overlap, while the majority of which carry modifiability, and that early intervention on these environmental factors contributes to preventing or alleviating NDDs (Bloem et al., 2021; Livingston et al., 2020; Scheltens et al., 2021; Thompson et al., 2018; van Es et al., 2017). However, some controversies linger among a large number of current investigations. In observational studies, the quality of evidence on interventions targeting environmental factors to decrease NDDs risk continues to be mixed, and the correlation between environmental factors with NDDs can be elusive to be interpreted as robust causal evidence due to potential confounding or reverse causation. Therefore, based on the causal associations between multiple exposures with NDDs remaining ambiguous, systematically disentangling their relationships is desirable for supporting the establishment of accurate risk evidence.

Mendelian randomization (MR) is a robust investigation method that uses genetic variants randomly allocated at conception as proxies for exposures and strengthens causal inference on exposure-outcome association (Davey Smith and Ebrahim, 2005; Davey Smith and Hemani, 2014). MR, as a study design located at a relatively higher hierarchy of the scientific evidence pyramid, can overcome some of the limitations of observational studies and is less susceptible to reverse causation and confounding factors, especially in scenarios where ethical and feasibility limitations render randomized controlled trials (RCTs) elusive. MR study design could provide reliable and valuable insights into the causal associations underlying the environmental factors and NDD (Arsenault, 2022; Haycock et al., 2016). Previous MR studies addressing environmental factors with NDDs have mostly focused on investigating the association between a single or a few exposures with a single outcome in NDDs with relative scattering findings due to data limitations (Andrews

et al., 2021; Domenighetti et al., 2022; Hu et al., 2022a, 2022b; Sproviero et al., 2021; Wu et al., 2021; Xia et al., 2023; Zeng and Zhou, 2019; Zhang et al., 2020), while an improvement in the integrality and systematicity of these studies is in demand. However, under the support of the increasing number of updated large-scale genome-wide association studies (GWASs), the breadth and depth of MR investigations into NDDs hold further potential for expansion.

Therefore, in this updated MR investigation, we adopted an environmental factor framework covering socioeconomic factors, behavioral factors, metabolic factors, and physical measurements as exposures, and major NDDs including AD, PD, ALS, and MS as outcomes, with the aims of 1) systematically investigating environmental factors with causal associations to NDDs, 2) integrally identifying the discrepancies in causal effects of the identical environmental factor on multiple NDDs, and 3) comprehensively revealing the differences in causal effects of multiple subgroups within the environmental factors on the identical NDD, in an effort to provide insights for establishing risk evidence for NDDs (Fig. 1).

2. Methods

2.1. Genome-wide association summary statistics of environmental factors

The 51 exposures were systematically categorized into four major groups of environmental factors, including socioeconomic factors, behavioral factors, metabolic factors, and physical measurements. The environmental factors with the largest sample sizes of relatively recent GWAS data were captured as exposures. Due to the extensive GWAS data that were included in this Mendelian randomization investigation, the details of each GWAS data including classification, sample size, main consortium/cohort, author (year), and PubMed ID were summarized in Table S6. The single nucleotide polymorphisms (SNPs) were extracted as instrumental variables (IVs) from summary statistics of the obtained GWASs of participants of European ancestry.

2.2. Genome-wide association summary statistics of neurodegenerative diseases

The summary statistics of the major NDDs including AD (Kunkle et al., 2019), PD (Nalls et al., 2019), ALS (van Rheenen et al., 2021), and MS (International Multiple Sclerosis Genetics, 2019) were extracted from four large-scale meta-GWASs of participants of European ancestry. These NDDs served as the outcomes of this MR investigation (Table S6).

The AD GWAS dataset from the International Genomics of Alzheimer's Project Stage 1 meta-analysis included 21,982 AD cases and 41,944 cognitively normal controls, incorporating 46 case-control studies from four consortia including the Alzheimer Disease Genetics Consortium (ADGC), the European Alzheimer's Disease Initiative (EADI), the Cohorts for Heart and Aging Research in Genomic Epidemiology Consortium (CHARGE), and the Genetic and Environmental Risk in AD/Defining Genetic, Polygenic and Environmental Risk for Alzheimer's Disease Consortium (GERAD/PERADES) (Kunkle et al., 2019). The PD GWAS dataset was obtained from a meta-GWAS released by the International Parkinson's Disease Genomics Consortium (IPDGC) and included 33,674 PD cases and 449,056 controls mainly from three previous studies and 13 case-control studies (Nalls et al., 2019). The ALS GWAS dataset was derived from a large-scale multi-source meta-GWAS based on individual-level genotype data from 117 cohorts, including 27,205 ALS cases and 110,881 controls (van Rheenen et al., 2021). The MS GWAS dataset was from a meta-GWAS published by the International Multiple Sclerosis Genetics Consortium (IMSGC) based on 15 previous datasets and two large-scale independent datasets, including a total of 47,429 MS cases and 68,374 controls (International Multiple Sclerosis Genetics, 2019).

The study design of the GWASs of NDDs referred to above, such as

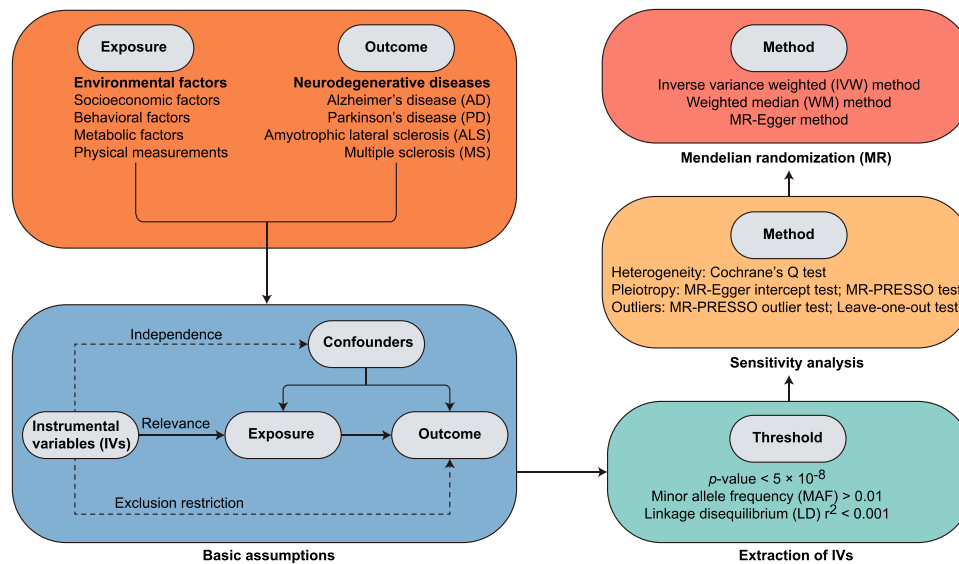


Fig. 1. : Overview of the study design. Flow diagram depicts this updated Mendelian randomization (MR) investigation on the causal associations of environmental factors on major neurodegenerative diseases (NDDs).

sample collection, quality control procedures, and imputation methods were described in the original publications, and ethical approval was obtained by each GWAS. The GWASs summary statistics of NDDs used in this two-sample MR investigation are the largest data to date, and we used data from different sources than the exposure GWASs to minimize sample overlap to ensure robustness.

2.3. Mendelian randomization study design

The MR analysis was performed in accordance with three basic assumptions: 1) the genetic variants are vigorously associated with the common environmental factors; 2) the genetic variants are independent of any potential confounders; and 3) the genetic variants influence NDDs only through each common environmental factors rather than other ways.

Preliminarily, SNPs with a p -value $< 5 \times 10^{-8}$ and minor allele frequency (MAF) > 0.01 were extracted. To obtain independent SNPs, linkage disequilibrium (LD) clumping was further performed by retaining SNPs that have an $r^2 < 0.001$ within a 10 MB window from the European panel in the 1000 genomes project using PLINK clumping method. Then, the F -statistic for each retained SNP was calculated to evaluate the SNPs' power (Table S6). The F -statistic estimates the strength of association between the genetic variants with the exposure, indicating the strength of the instrument. Concretely, the larger the F -statistic, the stronger the instrument and the higher the statistical power, while an F -statistic < 10 reflects a low instrument validity, then the corresponding SNPs would be removed. The F -statistic was calculated as follows (Burgess and Thompson, 2015):

$$F_{\text{statistic}} = \frac{\beta^2}{se^2} \quad (1)$$

Finally, the GWAS datasets of each set analysis of exposure and outcome were harmonized so that the effect size of both corresponded to the identical effect alleles. Genetic variants that were palindromic with ambiguous allele frequencies or that had incompatible alleles were removed. In addition, due to the large independent contribution of *APOE* $\epsilon 4$, a gene that has pleiotropy and influences multiple disease processes that contribute to the potential AD risk, which violates the basic MR basic assumptions, the SNPs rs7412 and rs429358 were removed when MR analysis was performed with AD as the outcome.

2.4. Statistical analysis

The Wald ratio was adopted to estimate the effect of each set of exposure on outcome for each IV, and then the effect size of each IV was combined using the inverse variance weighted (IVW) method to generate an overall estimation of the causal effect. In principle, the IVW method assumes that all SNPs involved in causal estimate do not violate any potential MR assumptions, and since the robustness of causal associations may be insufficient when any of the IVs has underlying effects on multiple phenotypes, relatively robust alternative MR methods including weighted median (WM) and MR-Egger were performed to complement any potential violation on MR assumptions by the IVW method. The WM method effectively pools the effects of individual variants under the principle that over 50% of the weight is derived from valid IVs, and the MR-Egger method allows all variants to be directly influenced, biasing the estimate in the same direction.

As a further sensitivity analysis to evaluate the causal association between each set of exposure and outcome, the leave-one-out analysis were performed to detect the outliers, the Cochrane's Q values were applied to assess the heterogeneity, and the MR-Egger intercept method was tested for pleiotropy. The MR Pleiotropy Residual Sum and Outlier (MR-PRESSO) test was performed to detect and correct for outliers with horizontal pleiotropy. The outliers would be removed and MR causal estimation would be reassessed when significant outliers are detected (MR-PRESSO outlier test).

All the statistical analyses and visualization were conducted using R version 4.2.1. The two-sample MR analysis was performed using the "TwoSampleMR" package, and the MR-PRESSO was performed using the "MRPRESSO" package. The heatmap was plotted using the "pheatmap" package, the forest plot using the "forestploter" package, and the scatter, funnel, and leave-one-out plots were directly generated from the "TwoSampleMR" package.

3. Results

In this MR investigation, we comprehensively examined the causal associations between the 51 exposures under the environmental factor framework with four major NDDs outcomes including AD, PD, ALS, and MS. Overall, our findings demonstrated that a total of 28 significant sets of exposure-outcome causal association evidence were identified. Of which, 12 sets of associations were previously underestimated and newly identified, including average weekly beer plus cider intake,

strenuous sports or other exercises (SSOE), diastolic blood pressure (DBP), and body fat percentage with AD, alcohol intake frequency with PD, apolipoprotein B, systolic blood pressure (SBP), and forced expiratory volume in 1 s (FEV₁) with ALS, and alcohol intake frequency, hip circumference, forced vital capacity (FVC), and FEV₁ with MS. Moreover, based on updated large-scale GWAS data and relatively rigorous methods, our MR investigation enhanced the robustness of some previously identified causal associations and disambiguated some associations that were overestimated in previous observational studies and MR studies. The number of SNPs involved in the environmental factors ranged from 3 to 498, and the *F*-statistic of each retained SNP was above the empirical threshold of 10, indicating that all the SNPs participating in this MR investigation had sufficient validity (Table S1-4; Table S6). The details of the findings of this MR investigation are shown as follows.

3.1. Socioeconomic factors and NDDs

Among the socioeconomic factors (Fig. 2A, B; Figs. S1-3; Table S5), higher educational attainment was predicted to be genetically associated with lower risk of AD (odds ratio [OR] = 0.707, confidence interval [CI] = 0.611–0.819, $p = 3.393 \times 10^{-6}$) and ALS (OR = 0.821, CI = 0.729–0.925, $p = 1.190 \times 10^{-3}$) (Fig. 2A, B), but higher risk of PD (OR = 1.466, CI = 1.208–1.780, $p = 1.112 \times 10^{-4}$) by the IVW method. Higher household income was causally associated with higher risk of PD (OR = 2.142, CI = 1.404–3.268, $p = 4.102 \times 10^{-4}$) (Fig. 2A, B). These causal effects were consistently supported by the WM method.

3.2. Behavioral factors and NDDs

Among the behavioral factors (Fig. 3A, B; Figs. S1-3; Table S5), in the subgroup of alcohol consumption, higher alcohol intake frequency was found to be causally associated with higher risk of MS (OR = 1.412, CI = 1.130–1.765, $p = 2.418 \times 10^{-3}$), but lower risk of PD (OR = 0.724, CI = 0.572–0.916, $p = 7.174 \times 10^{-3}$) by the IVW method. However, the causal effect of drinks per week with NDDs was not observed, and the presence of evidence of pleiotropy involving the causal association of alcohol intake frequency with MS suggested that this observed association may not be robust. Among the multiple types of alcohol consumption, higher average weekly beer plus cider intake, but not spirits or red wine, was causally associated with higher risk of AD (OR = 2.570, CI = 1.389–4.757, $p = 2.653 \times 10^{-3}$) (Fig. 3A, B).

In the subgroup of physical activity, moderate to vigorous physical activity (MVPA) was found to be causally associated with lower risk of MS (OR = 0.283, CI = 0.150–0.533, $p = 9.410 \times 10^{-5}$), while SSOE was causally associated with lower risk of AD (OR = 0.164, CI = 0.048–0.558, $p = 3.771 \times 10^{-3}$) (Fig. 3A, B) in comparison to not spending any time in SSOE.

In addition, the WM method revealed potential significant causal associations of ever smoked regularly with higher risk of PD (OR = 1.433, CI = 1.085–1.892, $p = 1.119 \times 10^{-2}$), higher cigarettes per day with higher risk of AD (OR = 1.197, CI = 1.023–1.400, $p = 2.508 \times 10^{-2}$), and high to infrequent or non-coffee consumers with higher risk of MS (OR = 1.151, CI = 1.004–1.321, $p = 4.383 \times 10^{-2}$), although these causal effects were not supported by the IVW method.

3.3. Metabolic factors and NDDs

Among the metabolic factors (Fig. 4A, B; Figs. S1-3; Table S5), in the subgroup of lipidemic traits, higher apolipoprotein B was predicted to be causally associated with higher risk of ALS (OR = 1.084, CI = 1.010–1.164, $p = 2.622 \times 10^{-2}$) by the IVW method. The potential significant causal association of higher apolipoprotein A-I with higher risk of MS (OR = 1.204, CI = 1.019–1.423, $p = 2.898 \times 10^{-2}$) was only supported by the WM method. In addition, the causal association between serum C-reactive protein (CRP) and AD turned from highly significant to insignificant after excluding the SNPs rs7412 and rs429358 of APOE ϵ 4.

3.4. Physical measurements and NDDs

Among the physical measurements (Fig. 5A, B; Figs. S1-3; Table S5), regarding the subgroup of blood pressure, higher SBP was predicted to be causally associated with higher risk of ALS (OR = 1.007, CI = 1.002–1.011, $p = 2.758 \times 10^{-3}$), and higher DBP was causally associated with lower risk of AD (OR = 0.987, CI = 0.976–0.997, $p = 1.145 \times 10^{-2}$) by the IVW method. In addition, higher hip circumference was found to be causally associated with lower risk of MS (OR = 0.576, CI = 0.428–0.775, $p = 2.760 \times 10^{-4}$), higher body fat percentage was causally associated with lower risk of AD (OR = 0.611, CI = 0.440–0.850, $p = 3.404 \times 10^{-3}$), and higher body mass index (BMI) was causally associated with higher risk of MS (OR = 1.365, CI = 1.201–1.552, $p = 1.956 \times 10^{-6}$) (Fig. 5A, B).

In the subgroup of lung function, higher FVC was found to be causally associated with lower risk of MS (OR = 0.827, CI = 0.698–0.979, $p = 2.743 \times 10^{-2}$), and higher FEV₁ was causally associated with lower risk of MS (OR = 0.816, CI = 0.699–0.952, $p = 9.589 \times 10^{-3}$), but higher risk of ALS (OR = 1.135, CI = 1.045–1.233, $p = 2.598 \times 10^{-3}$) (Fig. 5A, B).

Regarding the subgroup of cognition, better cognitive performance was predicted to be causally associated with lower risk of AD (OR = 0.753, CI = 0.667–0.849, $p = 3.895 \times 10^{-6}$) and ALS (OR = 0.843, CI = 0.765–0.930, $p = 6.412 \times 10^{-4}$), but higher risk of PD (OR = 1.343, CI = 1.142–1.580, $p = 3.697 \times 10^{-4}$) (Fig. 5A, B). Higher math class taken was causally associated with lower risk of AD (OR = 0.717, CI = 0.627–0.820, $p = 1.239 \times 10^{-6}$) and ALS (OR = 0.848, CI = 0.767–0.938,

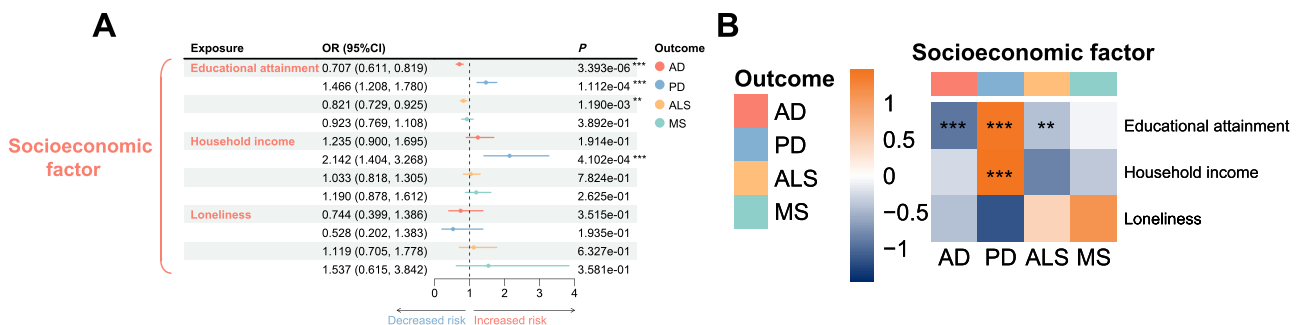


Fig. 2. : Causal associations of socioeconomic factors with major NDDs. (A) Forest plot depicts the causal associations of socioeconomic factors with AD, PD, ALS, and MS. The effect sizes are converted to OR with 95% CI. Significant causal associations are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). (B) Heatmap depicts the causal effects of socioeconomic factors on AD, PD, ALS, and MS. The standardized causal effect sizes are projected as color gradients, with values above or below zero representing increased or decreased risk of NDDs, respectively. Significant causal effects are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). AD, Alzheimer's disease; ALS, amyotrophic lateral sclerosis; CI, confidence interval; MS, multiple sclerosis; NDDs, neurodegenerative diseases; OR, odds ratio; PD, Parkinson's disease.

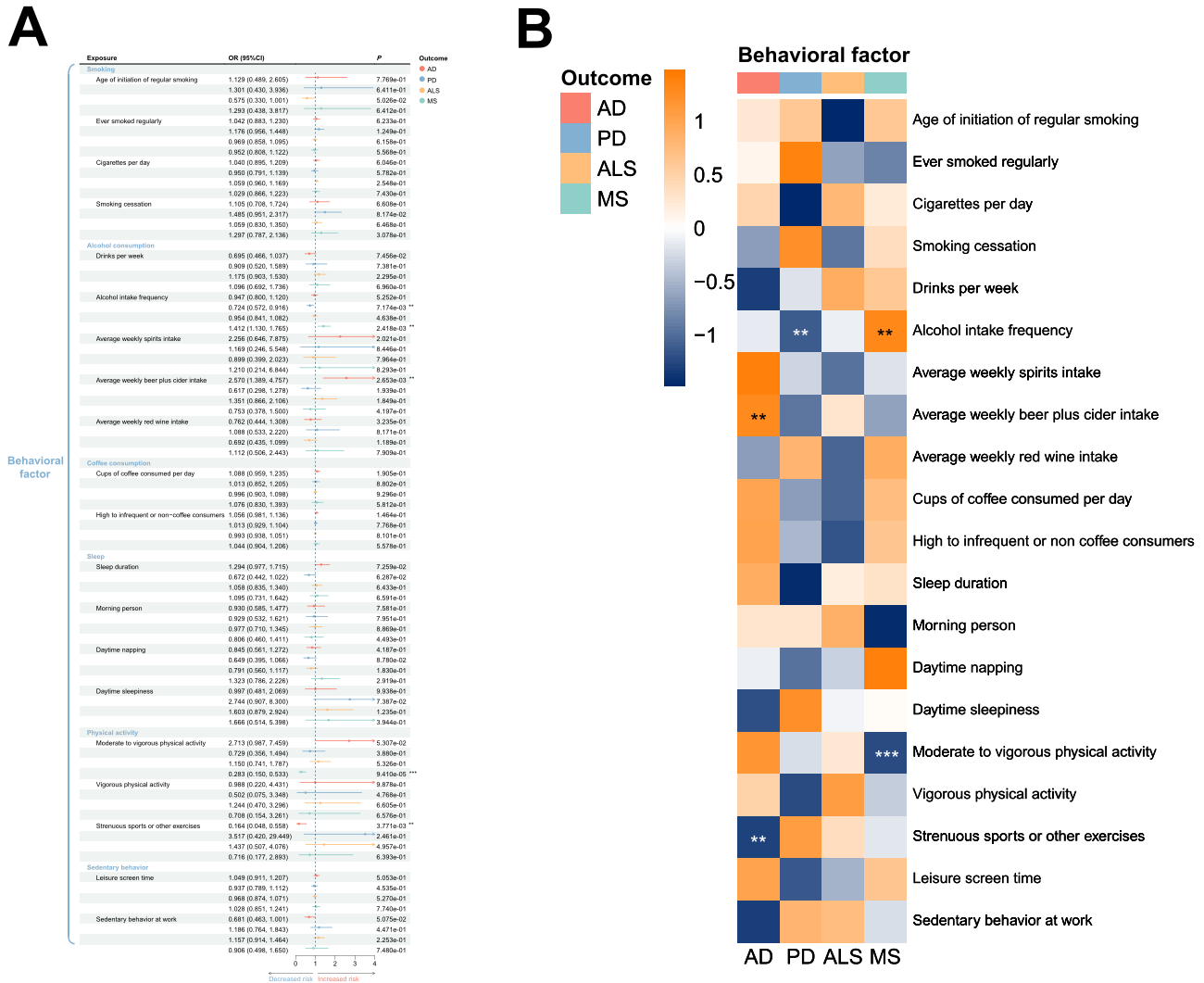


Fig. 3. : Causal associations of behavioral factors with major NDDs. (A) Forest plot depicts the causal associations of behavioral factors with AD, PD, ALS, and MS. The effect sizes are converted to OR with 95% CI. Significant causal associations are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). (B) Heatmap depicts the causal effects of behavioral factors on AD, PD, ALS, and MS. The standardized causal effect sizes are projected as color gradients, with values above or below zero representing increased or decreased risk of NDDs, respectively. Significant causal effects are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). AD, Alzheimer's disease; ALS, amyotrophic lateral sclerosis; CI, confidence interval; MS, multiple sclerosis; NDDs, neurodegenerative diseases; OR, odds ratio; PD, Parkinson's disease.

$p = 1.409 \times 10^{-3}$), but higher risk of PD (OR = 1.372, CI = 1.159–1.625, $p = 2.462 \times 10^{-4}$) (Fig. 5A, B). Better math ability was causally associated with lower risk of AD (OR = 0.784, CI = 0.677–0.908, $p = 1.193 \times 10^{-3}$), but higher risk of PD (OR = 1.301, CI = 1.085–1.559, $p = 4.412 \times 10^{-3}$) (Fig. 5A, B). While higher intelligence was predicted to be causally associated with lower risk of AD (OR = 0.703, CI = 0.603–0.820, $p = 6.858 \times 10^{-6}$) and ALS (OR = 0.864, CI = 0.767–0.973, $p = 1.630 \times 10^{-2}$) (Fig. 5A, B).

In addition, the WM method revealed potential significant causal associations between higher BMI and lower risk of AD (OR = 0.853, CI = 0.738–0.986, $p = 3.197 \times 10^{-2}$); however, the significance of this causal effect was not robustly supported by the IVW method.

3.5. Shared and mutually independent causal associations of environmental factors with NDDs

Overall, the causal associations of several environmental factors with NDDs overlapped (Fig. 6). Specifically, FEV₁, intelligence, alcohol intake frequency, and math ability were the causal environmental factors shared by ALS and MS, AD and ALS, PD and MS, and AD and PD,

respectively. Moreover, educational attainment, cognitive performance, and highest math class taken were shared by AD, PD, and ALS.

The causal effects of average weekly beer plus cider intake, DBP, body fat percentage, and SSOE on AD, household income on PD, apolipoprotein B and SBP on ALS, and BMI, FVC, hip circumference, and MVPA on MS were mutually independent.

4. Discussion

From an overall exposome-wide perspective, several shared causal associations of environmental factors with major NDDs were identified by 204 MR analyses, including FEV₁, intelligence, alcohol intake frequency, and math ability with ALS and MS, AD and ALS, PD and MS, and AD and PD, respectively. Moreover, educational attainment, cognitive performance, and highest math class taken were associated with AD, PD, and ALS jointly. Interestingly, within these shared causal associations, the causal effects of the identical environmental factors on diverse NDDs were not fully consistent. For example, obviously, our results suggested that several environmental factors including alcohol intake frequency, math ability, educational attainment, cognitive performance, and

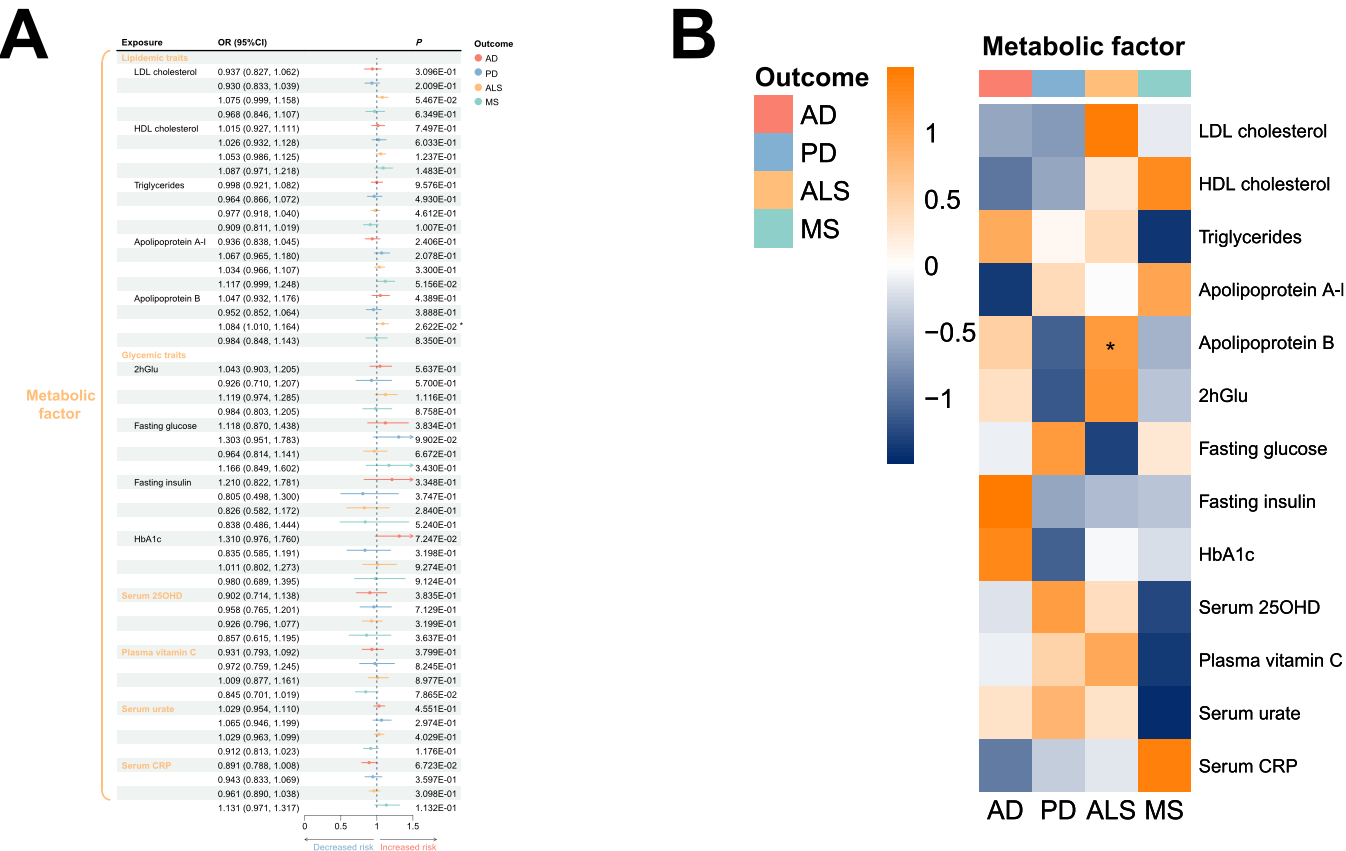


Fig. 4. : Causal associations of metabolic factors with major NDDs. (A) Forest plot depicts the causal associations of metabolic factors with AD, PD, ALS, and MS. The effect sizes are converted to OR with 95% CI. Significant causal associations are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). (B) Heatmap depicts the causal effects of metabolic factors on AD, PD, ALS, and MS. The standardized causal effect sizes are projected as color gradients, with values above or below zero representing increased or decreased risk of NDDs, respectively. Significant causal effects are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). AD, Alzheimer's disease; ALS, amyotrophic lateral sclerosis; CI, confidence interval; CRP, C-reactive protein; HbA1c, glycated hemoglobin; HDL, high-density lipoprotein; LDL, low-density lipoprotein; MS, multiple sclerosis; NDDs, neurodegenerative diseases; OR, odds ratio; PD, Parkinson's disease; 2hGlu, 2 h-glucose post-challenge; 25OHD, 25 hydroxyvitamin D.

highest math class taken possessed opposite causal effects on PD than their effect on other NDDs. Thus, the discrepancies among these causal associations provided the insight that some common prevention strategies may not be as generalizable as conventionally perceived. Significantly, we also found some mutually independent causal effects of environmental factors on NDDs, including average weekly beer plus cider intake, DBP, body fat percentage, and SSOE on AD, household income on PD, apolipoprotein B and SBP on ALS, and BMI, FVC, hip circumference, and MVPA on MS. These causal associations indicated that the prevention pathways across diverse NDDs may be partially specific, which provided a potential basis for establishing accurate risk evidence.

Among socioeconomic factors, a previous investigation indicated that low education in early life was a potential risk for AD (Livingston et al., 2020), which is further corroborated in our findings. Consistent with a previous MR analysis (Zhang et al., 2020), we also identified the causal effect of higher educational attainment to lower ALS. This causal inference is more robust due to a larger GWAS data of ALS and a rigorous clumping threshold in our study. Opposite to common perception, both higher educational attainment and household income were causally associated with higher PD risk. This finding is in accordance with the results from observational studies, which found that the association between higher educational attainment and the risk of PD may be linked to higher intelligence levels (Fardell et al., 2020), household income also potentially influences educational attainment (Haider and von Stumm, 2022; Rask-Andersen et al., 2021; von Stumm et al., 2020), and higher household income populations may receive more timely diagnosis

(Petersen et al., 2021). However, such causal associations need to be cautiously addressed, and the relationships of educational attainment and household income with PD remain to be further clarified in more extensive population-based studies.

Among behavioral factors, the evidence for the association of alcohol consumption with NDDs risk is inconclusive currently (Ascherio and Schwarzschild, 2016; Olsson et al., 2017). Our analysis demonstrated that, although the quantity of alcohol consumption did not establish causal relationships with major NDDs, higher alcohol intake frequency assessed from never, monthly, weekly, to daily showed that higher alcohol intake frequency was causally associated with a higher risk of MS and a lower risk of PD. Based on GWAS data with more adequate sample size, we complemented the previous MR analysis on the association of alcohol consumption with PD that was hampered by insufficient statistical power (Domenighetti et al., 2022). In addition, these findings were supported by the results from observational studies which suggested that alcohol consumption was a contributor to elevated MS risk and also in a negative, dose-dependent association with PD risk (Mitchell et al., 2022; Pakpoor et al., 2014; Zhang et al., 2014). On the aspect of the physical activity, we discovered a causal effect of SSOE for at least 2–3 days weekly or more on AD compared to no physical activity in the past four weeks, and a causal association of MVPA with lower MS risk. Multiple studies have also consistently indicated that physical activity benefits in maintaining cerebrovascular integrity, improving brain redox status, and increasing exercise-induced metabolic factors and muscle-derived myokines for neurogenesis, which promotes the production of neurotrophins and increases synapses and dendritic receptors

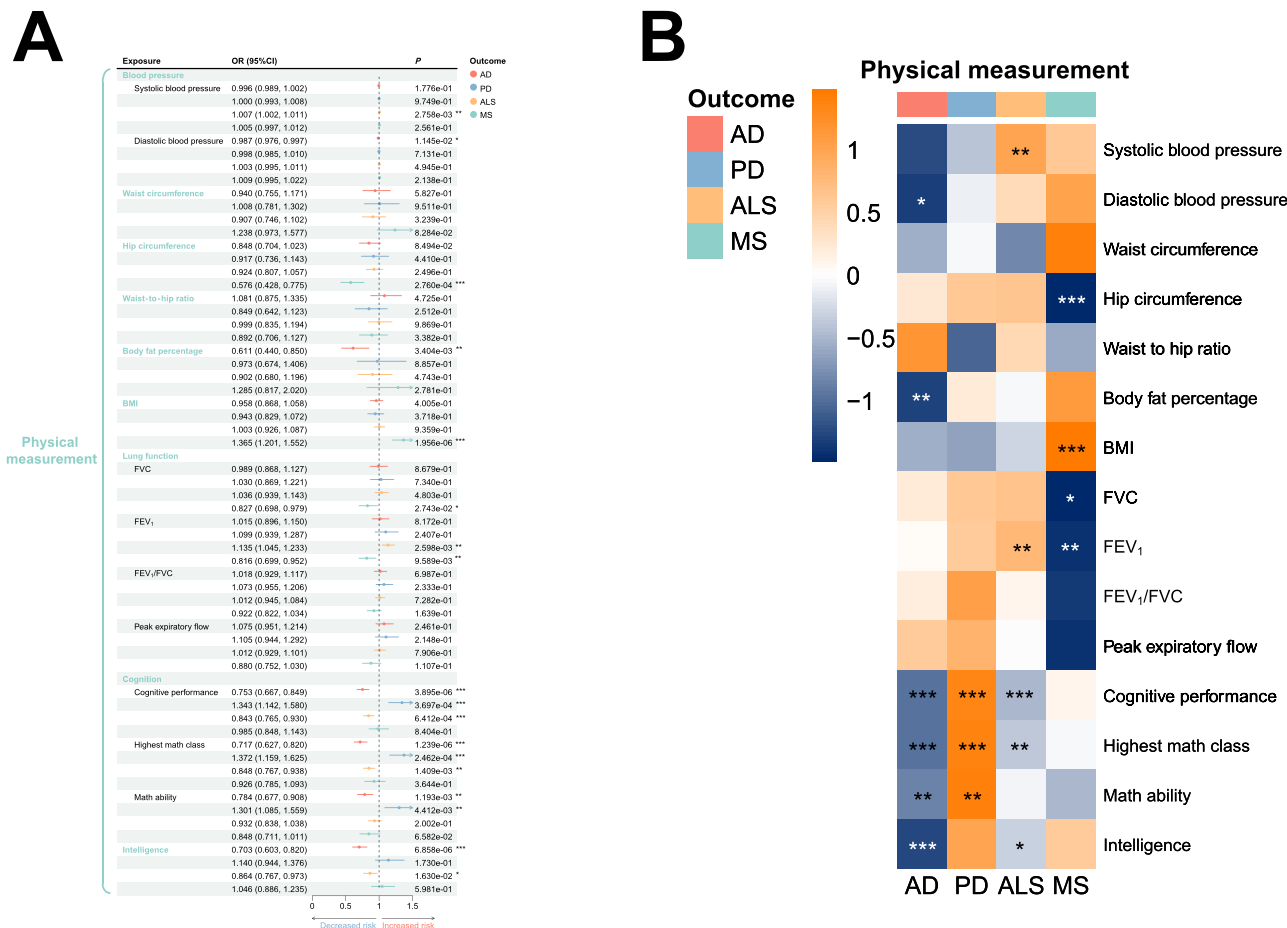


Fig. 5. : Causal associations of physical measurements with major NDDs. (A) Forest plot depicts the causal associations of physical measurements with AD, PD, ALS, and MS. The effect sizes are converted to OR with 95% CI. Significant causal associations are designated with asterisk (**p*-value < 0.05; ***p*-value < 0.01; ****p*-value < 0.001). (B) Heatmap depicts the causal effects of physical measurements on AD, PD, ALS, and MS. The standardized causal effect sizes are projected as color gradients, with values above or below zero representing increased or decreased risk of NDDs, respectively. Significant causal effects are designated with asterisk (**p*-value < 0.05; ***p*-value < 0.01; ****p*-value < 0.001). AD, Alzheimer's disease; ALS, amyotrophic lateral sclerosis; BMI, body mass index; CI, confidence interval; FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity; MS, multiple sclerosis; NDDs, neurodegenerative diseases; OR, odds ratio; PD, Parkinson's disease.

(Livingston et al., 2020; Sofi et al., 2011; Valenzuela et al., 2020; Wesnes et al., 2018).

Among metabolic factors, among multiple lipidemic traits, two previous MR analyses emphasized the causal effect of higher low-density lipoprotein (LDL) in increasing the ALS risk (Xia et al., 2023; Zeng and Zhou, 2019), however, we disambiguated this evidence based on larger exposure and ALS GWAS data. Only a significant causal association of higher apolipoprotein B with higher ALS was found and was consistent with the findings of cohort studies (Mariosa et al., 2017; Thompson et al., 2022). The alterations in apolipoprotein may be a precursor to the diagnosis of ALS decades before (Mariosa et al., 2017). Experimental results revealed that apolipoprotein B could potentially interfere with glia-neuron communication by modulating the activation of glial cells and that its higher levels could downregulate the vascular endothelial growth factor receptor 1 (VEGFR1) resulting in neuronal loss (Avraham-Davidi et al., 2012; Kunkle and Grenier-Boley, 2019). Currently, studies exploring the link between apolipoprotein A-I and MS remain insufficient, and we identified a causal association of higher apolipoprotein A-I on higher MS risk only with the support of the WM method, however, this conflicts with the findings from a limited number of observational studies (McComb et al., 2020; Murali et al., 2020). Further clarification of the association is pending. In addition, a previous MR analysis demonstrated a causal association of higher CRP with lower AD risk (Hu et al., 2022a). Although this finding is in agreement with evidence from observational studies (Hegazy et al., 2022), our replication

analysis suggested that the identified causal association was attributed to SNPs of *APOE* that have strong direct effects on AD, instead of being directly mediated through CRP, which violates the MR assumption. By removing the corresponding SNPs, we found that this causal association was not maintained, implying that the causal association of CRP to AD remains to be further examined.

Among physical measurements, the current studies have not converged on the relationship between BP and AD, possibly due to observational studies focusing on correlation instead of causation (Gregson et al., 2019; Ou et al., 2020). Our finding of the causal association between higher DBP with lower AD risk is consistent with a previous MR study that was independent of the AD GWAS data we used (Sproviero et al., 2021), but the causal effect of significant SBP on AD as in our results failed to be replicated. However, higher SBP rather than DBP showed a causal association with higher ALS risk. Evidence from a limited number of experiments indicated that abnormal sympathetic excitability could raise SBP (Aalkjaer et al., 2021), and exacerbate oxygen tension in ALS (Dalla Vecchia et al., 2015; Pimentel et al., 2019), leading to reductive bond cleavage and increased disordered superoxide dismutase 1 protein (SOD1) in patient cells (Keskin et al., 2019), potentially providing a mechanistic insight into the causal association of SBP with ALS. Furthermore, regarding obesity, we found a causal effect of higher hip circumference on lower MS risk and higher BMI demonstrated a significant causal association with higher MS risk. Previous investigations have consistently indicated that obesity in adolescence

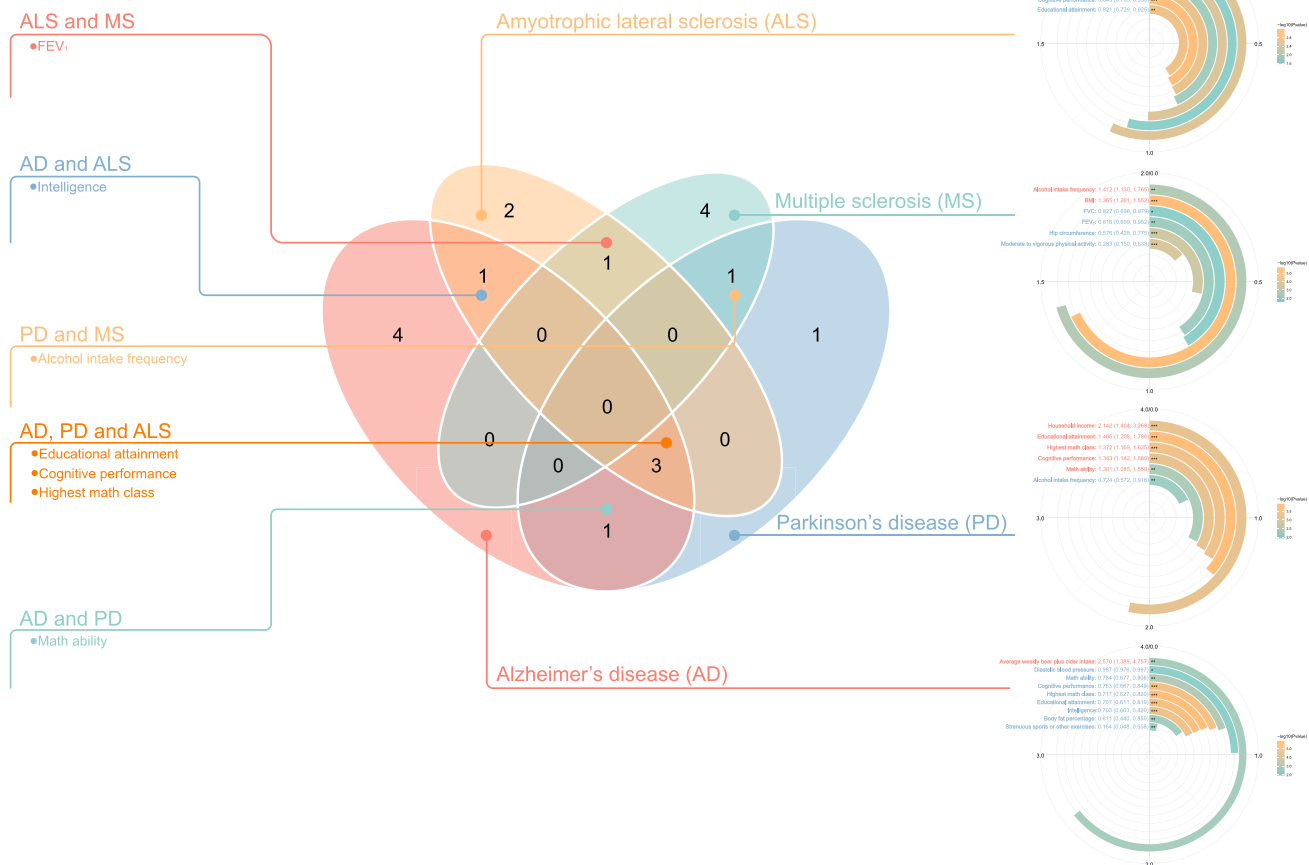


Fig. 6. : Shared and mutually independent causal associations of environmental factors with major NDDs. Venn diagram depicts the number of shared and mutually independent causal effects of environmental factors on AD, PD, ALS and MS, in which shared causal associations are extracted on the left. Radial histograms illustrate all significant causal associations that are identified, ORs are projected as lengths, p -values are projected as color gradients, and terms of environmental factors in red or blue represent increased or decreased risk of NDDs. Significant causal associations are designated with asterisk (* p -value < 0.05; ** p -value < 0.01; *** p -value < 0.001). AD, Alzheimer's disease; ALS, amyotrophic lateral sclerosis; BMI, body mass index; FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity; MS, multiple sclerosis; NDDs, neurodegenerative diseases; OR, odds ratio; PD, Parkinson's disease.

and possibly early adult life is a powerful risk factor for MS (Hegazy et al., 2022; International Multiple Sclerosis Genetics Consortium, 2019). Higher levels of the pro-inflammatory cytokine interleukin (IL)-6 and leptin in the cerebrospinal fluid of MS patients with high BMI suggested that inflammation could be a potential mediating mechanism involved in this causal association (Stampanoni Bassi et al., 2020). Regarding lung function, we identified causal associations of higher FVC with lower MS risk and higher FEV₁ with lower MS risk but higher ALS risk, however, there are limited studies in the literature review on these linkages. A case-control study revealed that FVC and FEV₁ were associated with disability in MS, and complications such as pneumonia resulting from the involvement of respiratory muscles that influence lung function measures were the most common cause of death in MS patients, which could be detected in the early stages of MS (Eren et al., 2021). Regarding cognition and intelligence, we found causal effects of better cognition and higher intelligence on lower AD risk. Results from RCTs suggested that cognitive interventions contribute to domain-specific improvements in cognitive function in the elderly (Ball et al., 2002; Willis et al., 2006), and a related MR study indicated a bidirectional causal effect between cognition and intelligence, and that both were independent of each other for the prevention of AD (Hu et al., 2022b).

However, several limitations of this MR investigation need to be

addressed. First, the possibility of violating independence and exclusion restriction assumptions cannot be completely avoided, but we applied sensitivity analysis methods including MR-Egger, leave-one-out, and MR-PRESSO to infer robust causal estimates to minimize the bias. Second, limited by the availability of GWAS data, some exposures potentially associated with NDDs from observational studies cannot be evaluated due to the absence of available IVs, and the probability of failure in detecting any linkage due to insufficient sample size cannot be ruled out. Third, GWAS data of NDDs are overall dichotomous, and large-scale GWAS data stratified by sex, age, NDD phenotypes, and NDD stages could contribute to further evidence of specific causal associations, additionally, the future public availability of more GWAS data unadjusted for inherited covariates would enhance the robustness of causal inferences. Fourth, although we comprehensively explored the causality of common environmental factors for the four major NDDs, limited by the lack of GWAS data on NDDs as a composite single outcome, the impact of multiple factors on integrated NDDs cannot be implemented, and future disclosure of integrated NDD-specific GWAS data would contribute to providing valuable insights.

Despite these limitations, our MR investigation holds several major strengths. First, this is an updated investigation using the latest and largest scale GWAS data of major NDDs to comprehensively evaluate the causal associations between multiple common exposures within the

environmental factor framework with major NDDs, and MR analysis strengthened the causal inference of the association of exposures with NDDs due to diminished confounding factors and reverse causality. Second, for all GWAS data, participants were of European ancestry, indicating that the population stratification is unlikely to bias our findings, rendering the investigation relatively robust, but expanding our findings to other populations should be addressed with caution. Third, we did not limit our investigation to a single phenotype for some environmental factors, but extended to multiple subgroups. Fourth, based on updated GWAS data and relatively rigorous but robust methods, we disambiguated or identified some causal effects of exposures on NDDs that were overestimated or underestimated in previous MR studies, and detected some discrepancies in the causal associations of the identical exposure on multiple NDDs or multiple exposures on identical NDD, providing data support for further studies in exploring the biological mechanisms of the differential effects of environmental factors on NDDs. In addition, with the gradual expansion of GWAS, the development of more robust methods to address issues like sample overlap correction, IVs screening process, and sensitivity analysis in the future would contribute to the improvement of the statistical precision of MR.

In conclusion, in this updated MR investigation, we extensively identified the causal associations of multiple exposures across socio-economic factors, behavioral factors, metabolic factors, and physical measurements under the framework of environmental factors with major NDDs including AD, PD, ALS, and MS, and revealed shared and mutually independent causal effects of environmental factors on diverse NDDs. From the perspective of the triangulation, the integration of findings across multiple study types to further minimize study bias is of significant public health implication in establishing robust evidence of causal associations between environmental factors and NDDs, providing informative support for primary prevention policies of NDDs.

Ethics approval and consent to participate

All GWAS data involved are de-identified and with public availability, and the ethical approval was obtained by each GWAS, which were described in the original publications.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.arr.2024.102254](https://doi.org/10.1016/j.arr.2024.102254).

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